

Perturbation Testing with Rate and Spiking RNNs: A Tutorial Reproduction with Learnable Time Constants

Perturbation Testing Tutorial

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Abstract

This note documents a minimal tutorial pipeline for *perturbation testing* of recurrent network models of cortical spontaneous activity, following Sourmpis et al. (eLife 2026). We fit data-constrained recurrent neural networks (RNNs) to Neuropixels recordings from a single Allen Institute session and ask whether the fitted models predict the effect of optogenetic stimulation of parvalbumin-positive (PV) inhibitory neurons — a perturbation the model never saw during training. We summarise the four network architectures (a vanilla rate RNN, a leaky integrate-and-fire spiking RNN, and their low-rank inter-area variants), the biological constraints that can be switched on (Dale’s law, E/I balance, spectral-radius control, inter-area Dale’s law), the training objective (a trial-averaged plus a trial-matched loss on an oscillation-feature embedding), and the evaluation metrics (PSTH correlation, Brain-FID, and trial-matched R^2). We additionally describe a **membrane time constant** for the low-rank models, parameterised so it stays positive and bounded, which can be either learned jointly with the weights or held fixed, and is exposed as a swept hyperparameter in the experiment driver.

1 Overview and goal

The aim of perturbation testing is to evaluate a model not on the data distribution it was trained on, but on its response to a causal intervention. Here the intervention is optogenetic excitation of PV interneurons. A spontaneous-activity generator is trained *only* on unperturbed gray-screen activity; at test time the recorded optogenetic light waveform $i(t)$ is injected into the model’s putative inhibitory units, and the model’s predicted population response is compared against the held-out optogenetic recordings. A model that has captured the right recurrent structure should generalise to the perturbation; one that has merely memorised the baseline statistics should not.

This is a minimal reproduction of Sourmpis et al., eLife 2026 [1], using data from the Allen Institute Visual Coding Neuropixels dataset [3], session 829720705 (*Pvalb-Cre* $\times$ *Ai32*, functional-connectivity protocol), which combines PV optogenetic stimulation with simultaneous Neuropixels recordings. The trial-matching training loss follows Sourmpis et al., NeurIPS 2023 [2].

2 Data

Spike trains from QC-passing units across six visual areas (VISp, VISr1, VISl, VISa1, VISpm, VISam) are binned at $dt = 10$ ms. Neurons are sorted so they are grouped by area, and every neuron carries an area label and a sign label.

Conditions. Trials are integer-coded: $c = 0$ for spontaneous (gray-screen) windows, $c \geq 1$ for drifting gratings, and $c = -1$ for optogenetic-perturbation trials. Training and the in-distribution test set use a 70/30 stratified split per condition; the optogenetic trials form a separate perturbation set. The generators are trained on the spontaneous block only ($c = 0$), binarised to spikes.

Neuron signs. Each unit is labelled excitatory (+1), inhibitory (−1), or outlier (0) from a fast-spiking waveform classifier. Optionally the “ideal” PV labels can instead be derived directly from the perturbation itself (units whose firing rises during the high-light phase of the strongest raised-cosine trials; flag `--ideal-pv-neurons`). Inhibitory/PV units are the ones *driven* by the opto light at test time; they are excluded from scoring (the light clamps them identically in data and model, so scoring them would inflate the metrics).

Perturbation waveforms. The Allen functional-connectivity protocol provides three light waveforms — `single_pulse`, `fast_pulses` (2.5 ms pulses at 10 Hz over 1 s), and `raised_cosine` — each at three intensity levels $\{1.3, 1.7, 2.0\}$. In-block `sham` segments (LED-free windows drawn from the gaps between light events) act as a matched control. Each perturbation trial is 1.5 s long with light onset at 0.5 s.

3 Models

All four models are *spontaneous-activity generators*: there is no external stimulus input; recurrent dynamics are driven only by per-neuron Gaussian noise. We write the recurrence as $\mathbf{u} = \mathbf{x}\mathbf{W}$, so W_{ij} is the weight from presynaptic i to postsynaptic j . All generators warm-start each call from the previous call’s detached end state (no backpropagation across calls).

3.1 Vanilla rate RNN (`rnn`)

A ReLU-like rate unit clipped to $[0, 1]$:

$$\mathbf{u}_t = \mathbf{x}_{t-1}\mathbf{W} + \mathbf{b} + \boldsymbol{\sigma} \odot \boldsymbol{\varepsilon}_t + \mathbf{p}_t, \quad (1)$$

$$\mathbf{x}_t = \text{clip}(\mathbf{u}_t, 0, 1), \quad (2)$$

where $\boldsymbol{\varepsilon}_t \sim \mathcal{N}(0, I)$, $\boldsymbol{\sigma}$ is a learnable per-neuron noise scale (initialised to 0.1 so the noise-driven network does not collapse to the zero fixed point), and $\mathbf{p}_t$ is the optional perturbation current (zero during training). The output is scaled by a learnable scalar.

3.2 Leaky integrate-and-fire spiking RNN (`lif`)

A spiking variant with a leaky membrane and per-synapse transmission delays. Each synapse $i \rightarrow j$ is assigned at initialisation to one of `num_delays` delay bins via a one-hot tensor $\mathbf{W}^{(d)}$, so it reads its presynaptic spike from its own delay in the past:

$$\mathbf{u}_t = \sum_k (\mathbf{z}_{t-k})(W_{..k}^{(d)} \odot \mathbf{W}) + \mathbf{b} + \boldsymbol{\sigma} \odot \boldsymbol{\varepsilon}_t + \mathbf{p}_t, \quad (3)$$

$$\mathbf{v}_t = \alpha \mathbf{v}_{t-1} + (1 - \alpha) \mathbf{u}_t - \mathbf{z}_{t-1} v_{\text{thr}}, \quad (4)$$

$$p_t = \sigma_{\text{logistic}}(\beta(\mathbf{v}_t - v_{\text{thr}})), \quad \mathbf{z}_t = \mathbb{K}[p_t > 0.5] + (p_t - \text{sg}(p_t)), \quad (5)$$

where α is the membrane leak/retention factor, v_{thr} the firing threshold, and the last term is a straight-through estimator (forward: sampled 0/1 spike; backward: gradient through $p_t = \sigma_{\text{logistic}}$). $\text{sg}(\cdot)$ denotes stop-gradient.

3.3 Low-rank inter-area rate RNN (**lowRNN**)

This model imposes *area structure* on the connectivity. The global weight matrix is assembled block-wise: each intra-area block is a full ($N_a \times N_a$) matrix, while each inter-area block is constrained to rank R :

$$\mathbf{W}^{(\text{tgt}, \text{src})} = \begin{cases} \mathbf{W}_{\text{full}} & \text{intra-area (tgt = src),} \\ MN^\top & \text{inter-area, unconstrained,} \\ \mathbf{s}_{\text{src}} \odot (MN^\top) & \text{inter-area, Dale-constrained,} \end{cases} \quad M \in \mathbb{R}^{N_{\text{src}} \times R}, N \in \mathbb{R}^{N_{\text{tgt}} \times R}. \quad (6)$$

The dense matrix has its diagonal zeroed. Crucially, the **lowRNN** dynamics are now *leaky*, with a learnable time constant (Section 5):

$$\mathbf{x}_t = \alpha \mathbf{x}_{t-1} + (1 - \alpha) \text{clip}(\mathbf{u}_t, 0, 1), \quad \mathbf{u}_t = \mathbf{x}_{t-1} \mathbf{W}_{\text{global}} + \mathbf{b} + \boldsymbol{\sigma} \odot \boldsymbol{\varepsilon}_t + \mathbf{p}_t. \quad (7)$$

Setting $\alpha \rightarrow 0$ recovers the memoryless map $\mathbf{x}_t = \text{clip}(\mathbf{u}_t)$ of the vanilla rate RNN; $\alpha \rightarrow 1$ makes the rate change arbitrarily slowly.

3.4 Low-rank spiking RNN (**lowBio**)

The biological counterpart of **lowRNN**: it combines the low-rank/area-block connectivity of Section 3.3 with the delayed-synapse LIF dynamics of **lif**. Its membrane leak α is the same learnable, time-constant-derived quantity used by **lowRNN**, so its membrane time constant is learned and swept as well.

4 Biological constraints

The constraints below are applied at initialisation and re-projected after every optimiser step (`apply_constraint`).

Dale’s law (sign constraint). With `--sign-constrained`, $\mathbf{W}$ is initialised sign-constrained and E/I-balanced per column (so the net presynaptic drive into each postsynaptic neuron is zero at init), and the signs are re-projected after each step (excitatory rows clamped ≥ 0 , inhibitory rows ≤ 0). The construction follows the recipe of [2].

Self-connection removal. The diagonal of $\mathbf{W}$ is held at zero.

Spectral-radius control. Because the recurrence has little leak, an expansive $\mathbf{W}$ would blow the dynamics up over a trial. After each step the spectral radius is re-pinned to at most 1.2 (the low-rank factors M, N are scaled symmetrically by $\sqrt{\cdot}$ so $MN^\top$ scales correctly).

Inter-area Dale’s law. For the low-rank models, `--inter-area-dale` additionally sign-constrains the inter-area blocks (the factors M, N are kept non-negative and the fixed row-sign vector $\mathbf{s}_{\text{src}}$ supplies the sign). Without it, intra-area blocks are sign-constrained but inter-area projections are free.

5 Membrane time constant: fixed or learnable

The membrane time constant τ of the low-rank models (`lowRNN` and `lowBio`) is parameterised in log-space so it is strictly positive for any real value, and the per-step retention/leak factor α used in Eqs. (7) is derived from it:

$$\tau = \exp(\theta), \quad \alpha = \exp(-1/\tau) \in (0, 1), \quad (8)$$

where θ is the underlying scalar. Larger τ gives α closer to 1, i.e. longer memory and slower dynamics. The initial value τ_0 is set by the `--tau-init` flag (units of time bins). Because $\alpha \in (0, 1)$ and $\tau > 0$ are guaranteed by construction, no extra projection is needed in `apply_constraint`.

The time constant can be either **learnable** or **fixed**:

- *Learnable* (default): θ is a trained parameter optimised jointly with the weights; the network learns its own integration timescale, initialised at τ_0 . The learned τ is logged each epoch.
- *Fixed* (`--fixed-tau`): θ is held constant as a non-trainable buffer, so $\tau \equiv \tau_0$ throughout training. Sweeping τ_0 in this mode performs a controlled scan over the time constant.

In the spiking `lowBio` model this same α is the membrane leak, so making it learnable directly learns (or, when fixed, prescribes) the neuronal integration timescale.

6 Training objective

Models are trained to match recorded spontaneous activity through two complementary losses computed on low-pass-filtered rasters (a length-5 moving average), then combined with coefficients λ_{ta} and λ_{tm} :

$$\mathcal{L} = \lambda_{ta} \mathcal{L}_{ta} + \lambda_{tm} \mathcal{L}_{tm}. \quad (9)$$

Trial-averaged loss $\mathcal{L}_{ta}$. The squared difference between the trial-averaged (PSTH) low-passed activity of generated and recorded batches. This roughly corresponds to the paper’s per-neuron term.

Trial-matched loss $\mathcal{L}_{tm}$. Single trials carry variability that a PSTH loss ignores. Following [2], generated and recorded trials are embedded with a differentiable feature map $\phi(\cdot)$, an optimal one-to-one assignment between the two sets of trials is solved on the *detached* cost matrix (Hungarian algorithm), and the MSE over matched pairs is minimised (gradient flows through the matched feature entries).

Feature map ϕ . Both the differentiable loss (torch) and the numpy evaluation share a single feature pipeline so they cannot drift apart: (i) average activity within each area to obtain per-area population channels; (ii) project each channel onto a bank of log-spaced cosine/sine oscillation bands and take per-band amplitudes plus a DC term; (iii) z-score against the training-set statistics. Optimisation uses AdamW (learning rate 10^{-3} , weight decay 0.1); `apply_constraint` runs after each step. This is the “sample-and-measure” paradigm of [4]: generate freely, then penalise the dissimilarity between summary statistics of generated and recorded activity.

7 Theoretical background and related work

This section explains *why* the objective takes the form of Section 6, and in particular why the generators are trained free-running rather than autoregressively.

7.1 Generative distribution matching, not likelihood

The networks are *generative* models of spontaneous activity: noise is the only drive, and each **generate** call produces statistically independent trials. The training target is therefore the *distribution* of activity, not the next spike given the recorded past. The classical alternative — maximising the likelihood of the recorded spikes, as generalised linear models (GLMs) do — clamps the network’s trajectory to the data and optimises one-step-ahead conditionals. As [4] showed, such likelihood-fit recurrent spiking networks “do not produce realistic neural activity” when run autonomously. The sample-and-measure approach instead back-propagates through stochastically simulated spike trains to match neuroscience summary statistics of the *freely generated* activity, which is exactly the regime used at test time.

7.2 Why two complementary losses

The two terms constrain different statistical moments and neither suffices alone:

- $\mathcal{L}_{\text{ta}}$ (the PSTH/ $\mathcal{L}_{\text{neuron}}$ term) pins down *each neuron’s* trial-averaged response — mean rate and time course. But because it averages over trials, a degenerate model that emits the mean PSTH on every trial (zero variability) scores perfectly; the loss is blind to trial-to-trial structure. [2] show PSTH-only models cannot reproduce, e.g., a bimodal single-trial distribution.
- $\mathcal{L}_{\text{tm}}$ (the trial-matching term) pins down the *distribution over single trials*. Since generated and recorded trials are unordered and independent (“no reason for generated trial k to correspond to recorded trial k ”), it uses optimal transport: an assignment between the two trial sets in feature space (Hungarian algorithm on a *detached* cost, gradient through the matched entries) followed by a matched MSE. But on its own, the population-level features under-constrain individual neurons.

Together they yield correct single-neuron means *and* correct population trial-to-trial covariability. The eval-time Brain-FID (Section 9) is the non-differentiable, distribution-level analogue of $\mathcal{L}_{\text{tm}}$ on the same feature map.

7.3 Why free-running, not autoregressive / teacher-forced

Teacher forcing (feeding the recorded past as input) would be the autoregressive alternative. It is deliberately avoided here, for four reasons:

1. **Exposure bias.** A teacher-forced model conditions on *true* past states during training but on its *own* states at generation; the free-running dynamics it must ultimately produce are never directly optimised and can drift or destabilise. This train–test mismatch is the long-known motivation for fixes such as Professor Forcing [6] and scheduled sampling. Sample-and-measure sidesteps it by optimising the autonomous dynamics directly.
2. **The variability is the signal.** For spontaneous, noise-driven data there is no stimulus to “predict from”. Teacher forcing would let the network lean on the clamped recorded past instead of learning the recurrent structure that *generates* the observed variability — which is precisely the scientific object of interest.

3. **Perturbation testing requires autonomy.** The end goal is a counterfactual: inject the optogenetic current into PV units and predict the response. In that counterfactual there are *no* recorded spikes to clamp to, so the model must run autonomously. A teacher-forced model has no validated free-running behaviour and cannot be used for this prediction.
4. **Hidden neurons.** Clamping requires observing every neuron; sample-and-measure handles unobserved/hidden units naturally [4].

Note the code still uses back-propagation *within* a trial; it merely warm-starts each call from the *detached* previous end state (`self._state = x.detach()`) and never conditions on recorded spikes, so there is no gradient coupling across calls and no teacher forcing.

7.4 Related work

GLM and maximum-likelihood recurrent spiking networks fit one-step conditionals with clamped trajectories and are the teacher-forced baseline criticised above. [4] introduced the free-running sample-and-measure paradigm; [2] added the optimal-transport trial-matching loss to capture variability; the present pipeline follows [1], which applies these models to optogenetic perturbation prediction. A different lineage — sequential variational auto-encoders such as LFADS [5] — uses amortised inference to recover per-trial initial conditions and inferred inputs and *reconstructs* observed trials; it excels at single-trial denoising but is not a pure free-running generator matched at the distribution level, and its inferred inputs complicate clean perturbation counterfactuals. Distribution-level generative models of spikes (GANs, latent diffusion) share the “match the distribution” goal with very different, less mechanistic machinery.

8 Autoregressive training mode (alternative objective)

As a deliberate counterpoint to the free-running, distribution-matching objective of Section 6, the pipeline also offers an **autoregressive roll-out** objective, selected with `--train-mode autoregressive` (the default remains `trialmatch`). Here the network state is initialised *from data* and rolled out open-loop, and the prediction is matched to the *same* trials — so trial correspondence is given by construction and no optimal-transport assignment is needed.

8.1 Procedure

For a recorded trial $\mathbf{z}^{\mathcal{D}}$ and a start bin t_0 , the state is seeded from the data and the model free-runs for R steps (`--rollout-len`), producing $\hat{\mathbf{z}}_{t_0:t_0+R}$:

- *Rate models* (`rnn`, `lowRNN`): the rate state is set to the last observed bin, $\mathbf{x}_{t_0-1} = \mathbf{z}_{t_0-1}^{\mathcal{D}}$.
- *Spiking models* (`lif`, `lowBio`): the delayed-spike buffer is filled with the last `num_delays` recorded bins and the membrane is started at $\mathbf{v} = 0$ (an approximation; the first steps are a transient).

Several start bins t_0 are sampled *within* each trial per batch (`--n-inits`), yielding many initial conditions. The roll-out is open-loop (the network feeds back its own output — no teacher forcing during the roll-out); only the initial condition comes from data. As in the free-running mode, gradients flow within the roll-out but the persistent warm-start buffer is left untouched.

8.2 Losses

The trial-matching term $\mathcal{L}_{\text{tm}}$ is dropped. The objective is

$$\mathcal{L} = \lambda_{\text{ta}} \mathcal{L}_{\text{ta}} + \lambda_{\text{mse}} \mathcal{L}_{\text{roll}} + \lambda_{\text{band}} \mathcal{L}_{\text{band}}, \quad (10)$$

where (i) $\mathcal{L}_{\text{ta}}$ is the same free-running trial-averaged PSTH loss as before (it still constrains global per-neuron means); (ii) $\mathcal{L}_{\text{roll}}$ is the element-wise MSE between the low-pass-filtered roll-out and the matched data segment, $\|\text{lp}(\hat{\mathbf{z}}) - \text{lp}(\mathbf{z}_{t_0:t_0+R}^{\mathcal{D}})\|^2$; and (iii) $\mathcal{L}_{\text{band}}$ is the MSE between the oscillation-band features ϕ of the roll-out and of the *same* data segment — the same embedding as $\mathcal{L}_{\text{tm}}$ but with the trials matched by construction (no Hungarian step). Coefficients are `--coeff_ta`, `--coeff_ar_mse`, `--coeff_ar_band`. Because the roll-out is stochastic, the smoothing keeps $\mathcal{L}_{\text{roll}}$ meaningful only for the low-frequency, short-horizon component; the band term carries the oscillation content over the window. Since the features are integrated over the R -bin window, their normalisation statistics are refit on R -length crops of the training data so the lowest bands remain comparable.

8.3 Effect on metrics and plots

The free-running generative metrics of Section 9 (PSTH- r , Brain-FID, optimal-transport R^2) are still computed each epoch. In addition, a **matched roll-out** evaluation is reported: test trials are seeded from data, rolled out for R steps, and scored against the matched data segment with the trial order preserved (the `matched=True` path), giving a matched roll-out R^2 that becomes the primary tracked quantity and adds a fourth panel to the over-epochs figure. The training raster switches from free-running samples to a *matched prediction*: each data trial is shown beside its own roll-out (data prefix followed by the open-loop continuation). The perturbation-testing evaluation is **unchanged** — it remains free-running with the injected opto current, since the counterfactual has no spikes to seed from (Section 7).

9 Evaluation metrics

Three metrics compare model-generated rasters to held-out recorded rasters of the same shape:

1. **PSTH Pearson r** ($\uparrow$): per-neuron Pearson correlation between trial-averaged firing rates (model vs. data). Trials need not be matched.
2. **Brain-FID** ($\downarrow$): Fréchet distance between Gaussian fits to the distribution of single-trial feature vectors $\phi(z)$ of model vs. data (a distribution-level score, in the spirit of the image FID).
3. **Trial-matched R^2** ($\uparrow$): R^2 between optimally matched single-trial populations on the same feature embedding ϕ used by Brain-FID.

Two reference levels frame the scores: *chance baselines* from time-shuffled and neuron-shuffled rasters, and an *ideal ceiling* from comparing real train data against real test data. Brain-FID and trial-matched R^2 deliberately share the feature map ϕ so the two numbers are consistent.

Perturbation evaluation. After training, the recorded light waveform $i(t)$ for each condition is injected as a perturbation current into the driven (PV/inhibitory) units, scaled by an *opto intensity* factor that is swept over `--opto-intensities`. Metrics are computed on the non-driven units only. Conditions span `sham` and every (waveform, level) combination present in the session.

Flag	Meaning	Default
--model	rnn / lif / lowRNN / lowBio	rnn
--sign-constrained	enable Dale's law (sign-constrained, E/I-balanced $\mathbf{W}$)	off
--inter-area-dale	sign-constrain inter-area blocks (low-rank models)	off
--rank	inter-area low-rank R (low-rank models)	2
--tau-init	initial/fixed membrane time constant (bins)	2.0
--fixed-tau	hold τ fixed at --tau-init instead of learning it	off (learnable)
--num-delays	number of synaptic delay bins (spiking models)	3
--ideal-pv-neurons	use perturbation-derived PV labels instead of waveform	off
--epochs	training epochs	20
--batch-size	trials per batch	64
--lr	AdamW learning rate	10^{-3}
--coeff_ta	weight λ_{ta} of the trial-averaged loss	1.0
--coeff_tm	weight λ_{tm} of the trial-matched loss	0.3
--train-mode	trialmatch / autoregressive (Section 8)	trialmatch
--rollout-len	roll-out length R in bins (autoregressive)	50
--n-inits	within-trial initial conditions per batch (autoregressive)	4
--coeff_ar_mse	weight λ_{mse} of the roll-out MSE (autoregressive)	1.0
--coeff_ar_band	weight λ_{band} of the matched-band MSE (autoregressive)	0.3
--opto-intensities	perturbation-current scale factors (swept at eval)	0.1 0.5 1.0

Table 1: Command-line settings of the training script. The time constant (--tau-init) is the newly added learnable, swept parameter.

Model family	Swept axes	Runs
rnn	{unconstrained, sign-constrained}	2
lif	{unconstrained, sign-constrained}	2
lowRNN, lowBio	{unconstr., sign-constr., inter-Dale} $\times$ rank{1, 2, 3, 4} $\times$ τ_0 {1, 2, 4, 8}	$2 \times 3 \times 4 \times 4 = 96$
Total		100

Table 2: Sweep performed by run_baselines.sh. The low-rank families now sweep the time-constant initialisation τ_0 in addition to rank and the constraint configuration. Each run additionally evaluates every opto-intensity in --opto-intensities. Results are written to figures/<model>_<constraint>[_rank<R>][_tau<T>]/.

10 Settings and experiment sweep

Table 1 lists the command-line settings of train_rnn.py. Table 2 summarises the full sweep run by run_baselines.sh.

11 Results

We report results from the sweep of Section 10, evaluated on the held-out in-distribution test set (free-running spontaneous activity) and on the held-out optogenetic perturbation set. All figures are produced directly by the experiment driver and live under figure_experiments/; the per-run metrics are aggregated in metrics_summary.csv (free-running and matched roll-out scores) and perturbation_metrics.csv (per-condition perturbation scores). Unless noted, the qualitative panels use the best-scoring free-running configuration, lowBio with Dale's law, rank 2 and $\tau_0 = 2$.

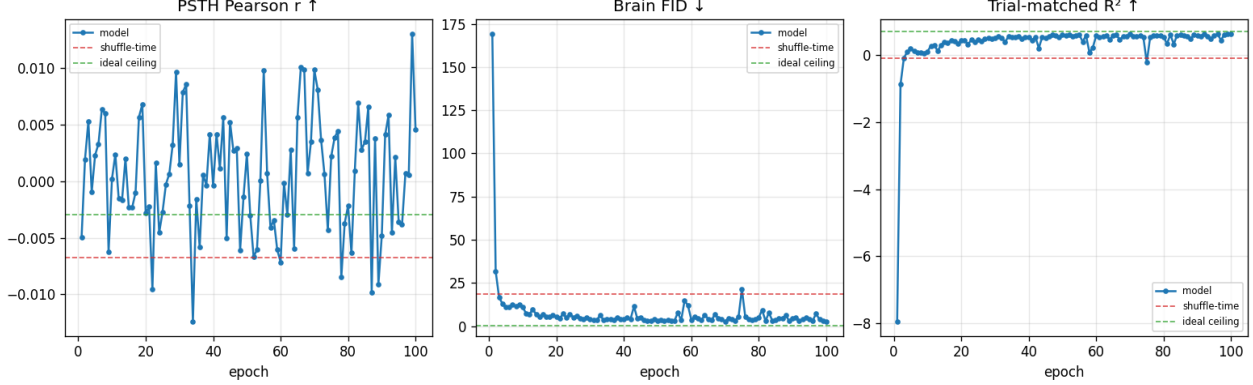


Figure 1: Free-running metrics over training for `lowBio`, Dale’s law, rank 2, $\tau_0 = 2$. Left: PSTH Pearson r ; middle: Brain-FID ($\downarrow$); right: trial-matched R^2 ($\uparrow$). Dashed red is the time-shuffle chance level, dashed green the real-train vs. real-test ideal ceiling.

11.1 Training converges on the free-running metrics

Figure 1 shows the three free-running metrics over training for a representative low-rank spiking model. The Brain-FID drops from ~ 170 at initialisation to below the time-shuffle chance level within a few epochs and settles near the ideal ceiling (real-train vs. real-test), while the trial-matched R^2 rises above chance to ≈ 0.6 . The PSTH Pearson r stays near zero throughout: with noise-driven generators and no stimulus, the per-neuron trial-averaged trace carries little signal, so the distribution-level scores (Brain-FID, trial- R^2) are the informative ones — exactly the motivation for the sample-and-measure objective of Section 7.

11.2 Free-running fit across models

Table 3 summarises the free-running test scores for one representative configuration per model family (best τ_0 /rank for the low-rank models). Two patterns are robust across the sweep. First, the area-structured low-rank models (`lowRNN`, `lowBio`) reach the lowest Brain-FID and the highest trial- R^2 , with the spiking `lowBio` best overall (Brain-FID ≈ 2.8 , trial- $R^2 \approx 0.63$), well inside the chance–ceiling band. Second, the free-running, distribution-matching objective decisively beats the autoregressive roll-out objective on these distribution-level scores: in autoregressive mode the free-running Brain-FID and trial- R^2 collapse (most configurations score negative trial- R^2 ; cf. the `ar` rows of `metrics_summary.csv`), consistent with the exposure-bias argument of Section 7. The matched roll-out R^2 that the autoregressive mode optimises stays modest ($\lesssim 0.16$), i.e. open-loop prediction of single spontaneous trials remains hard even when trials are matched by construction.

11.3 Perturbation testing

The decisive test is the held-out optogenetic perturbation, which no model saw during training. Figures 2 and 3 sweep the raised-cosine perturbation: each panel plots a metric against the recorded light intensity (1.3, 1.7, 2.0), with one row per training mode (`freerun/ar`) and one column per opto-injection scale (`--opto-intensities` $\in \{0.01, 0.1, 1.0\}$); lines are means over rank and τ_0 per model family, and the dashed traces are the no-perturbation baseline. Performance degrades markedly relative to the free-running fit: Brain-FID rises well above its unperturbed value and trial- R^2 frequently goes negative, i.e. predicting the causal effect of PV stimulation is substantially harder than matching spontaneous statistics — the expected and scientifically interesting outcome

Model	Constraint	rank, τ_0	Brain-FID $\downarrow$	trial- R^2 $\uparrow$
rnn	unconstrained	—	8.41	0.458
rnn	Dale	—	14.71	0.165
lif	unconstrained	—	8.94	0.355
lif	Dale	—	6.92	0.501
lowRNN	Dale	2, 2	4.52	0.609
lowBio	Dale	2, 2	2.79	0.630

Table 3: Free-running test metrics (**freerun** mode) for representative configurations. Lower Brain-FID and higher trial- R^2 are better. The area-structured low-rank models dominate, and the spiking **lowBio** is best.

of perturbation testing. The per-run heatmap of Figure 4 shows this is consistent across the full sweep (columns are opto-scale $\times$ light-strength), with the low-rank spiking models again the least degraded.

11.4 Qualitative perturbation response

Figure 5 shows ground-truth (left) vs. generated (right) rasters for the strongest raised-cosine condition. The model’s trial-averaged population activity (top panels) clearly tracks the injected raised-cosine light waveform, so the network does produce a perturbation-locked response of roughly the right gross shape; but the single-trial structure (lower panels) and the amplitude/timing details diverge from the data, which is why the quantitative trial- R^2 for this condition is poor despite the qualitatively sensible drive. This gap between a plausible mean response and a correct single-trial distribution is precisely what the distribution-level perturbation metrics are designed to expose.

12 Conclusion

The tutorial provides a compact, end-to-end perturbation-testing pipeline: data-constrained rate and spiking RNNs, optional biological constraints, a trial-matching training objective, and distribution-level metrics evaluated under a held-out optogenetic perturbation. The low-rank, area-structured models (**lowRNN**, **lowBio**) now carry a learnable membrane time constant that is both fit by gradient descent and swept across initialisations, letting the neuronal integration timescale be studied as a first-class hyperparameter alongside connectivity rank and Dale’s-law constraints.

References

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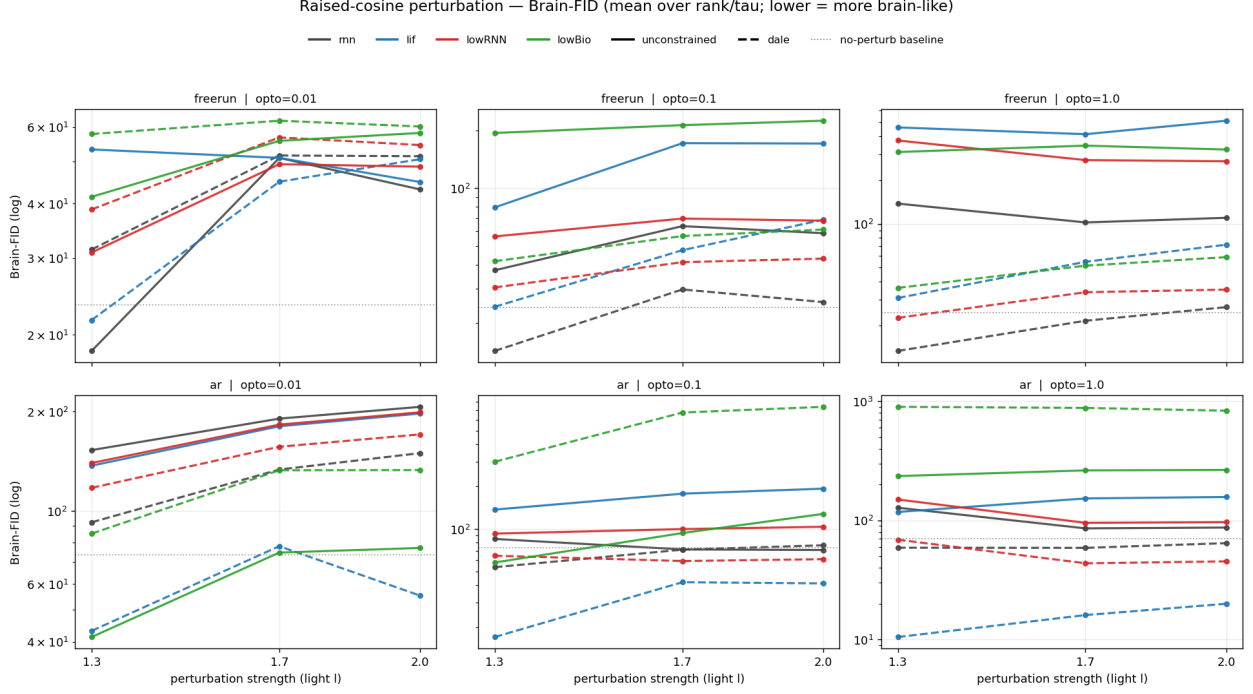


Figure 2: Raised-cosine perturbation, Brain-FID ($\downarrow$) vs. light strength. Rows: training mode (**freerun**, top; **ar**, bottom). Columns: opto injection scale. Solid lines are model families (mean over rank/ τ_0); dashed lines are the no-perturbation baseline.

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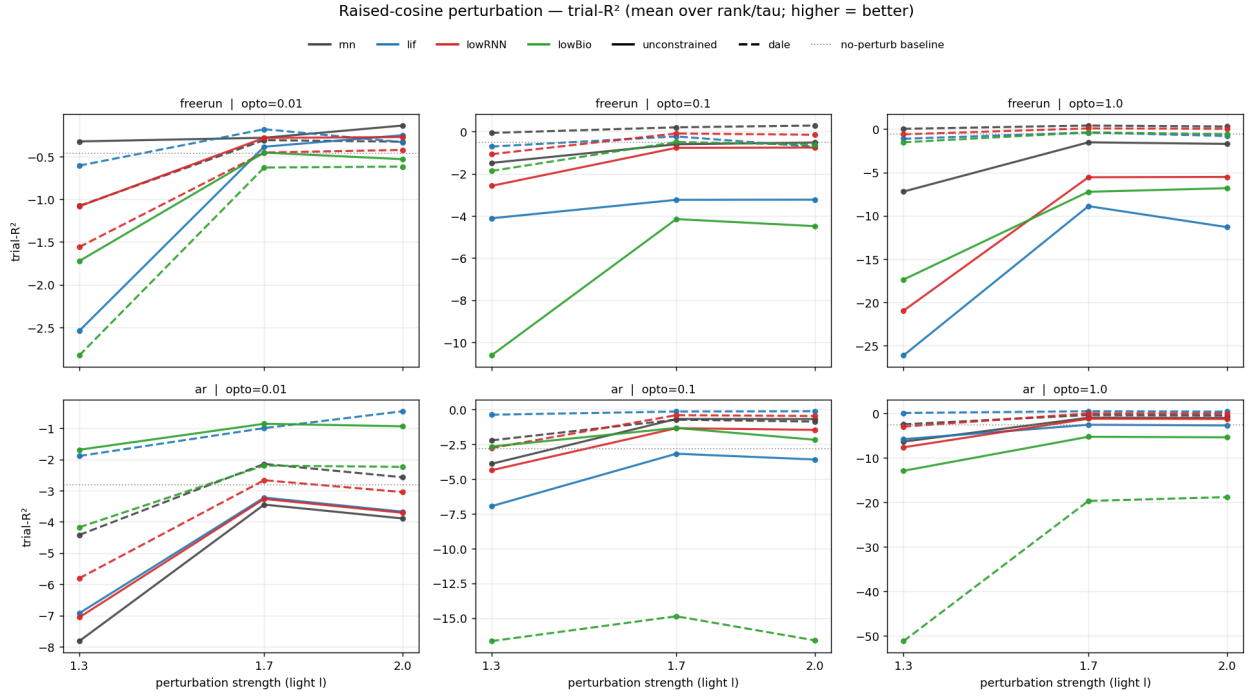


Figure 3: Raised-cosine perturbation, trial-matched R^2 ($\uparrow$) vs. light strength, same layout as Figure 2. Values frequently fall below zero, quantifying how much harder the held-out perturbation is than the free-running fit.

Raised-cosine perturbation — Brain-FID per run (columns = opto $\times$ light strength)

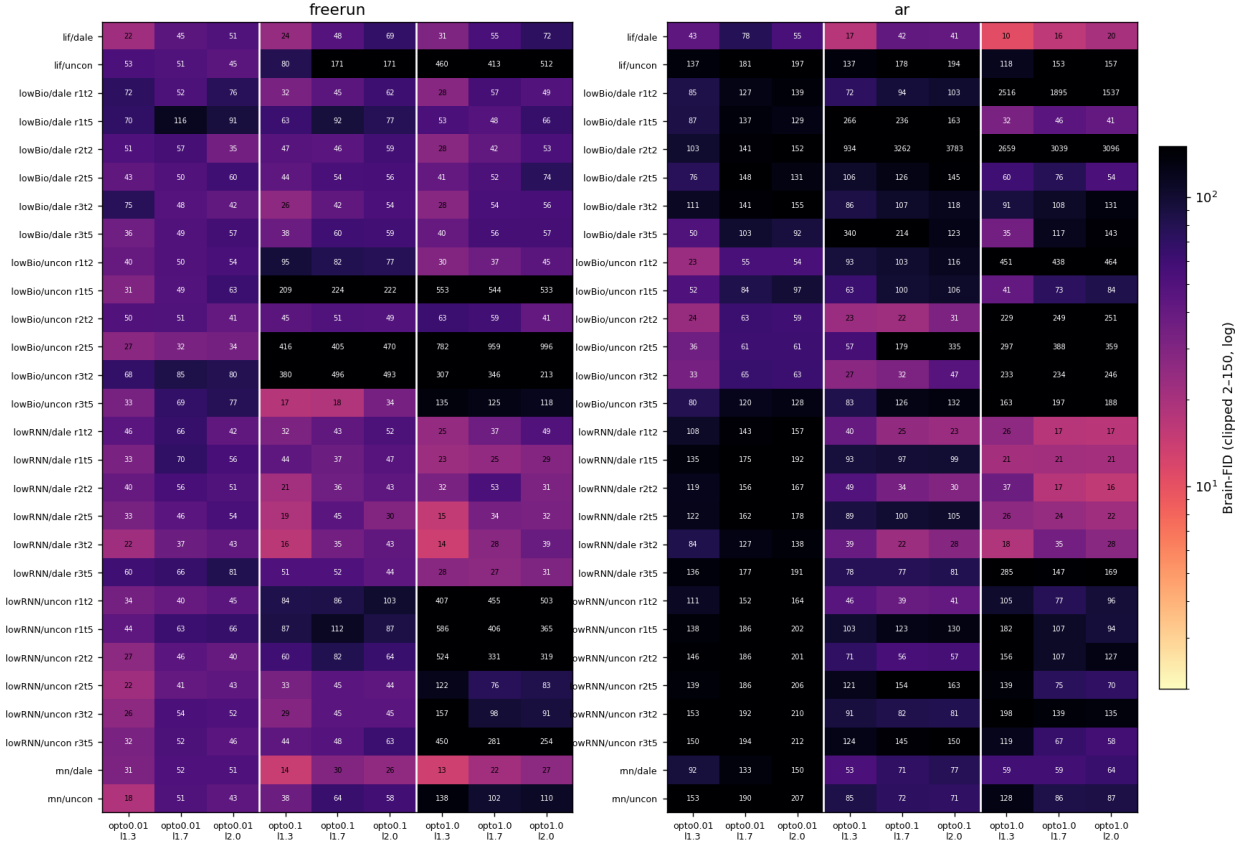


Figure 4: Per-run Brain-FID under the raised-cosine perturbation (log colour scale). Rows are individual runs (model $\times$ constraint $\times$ rank $\times \tau_0$), columns are opto-scale $\times$ light-strength; **freerun** (left) vs. **ar** (right).

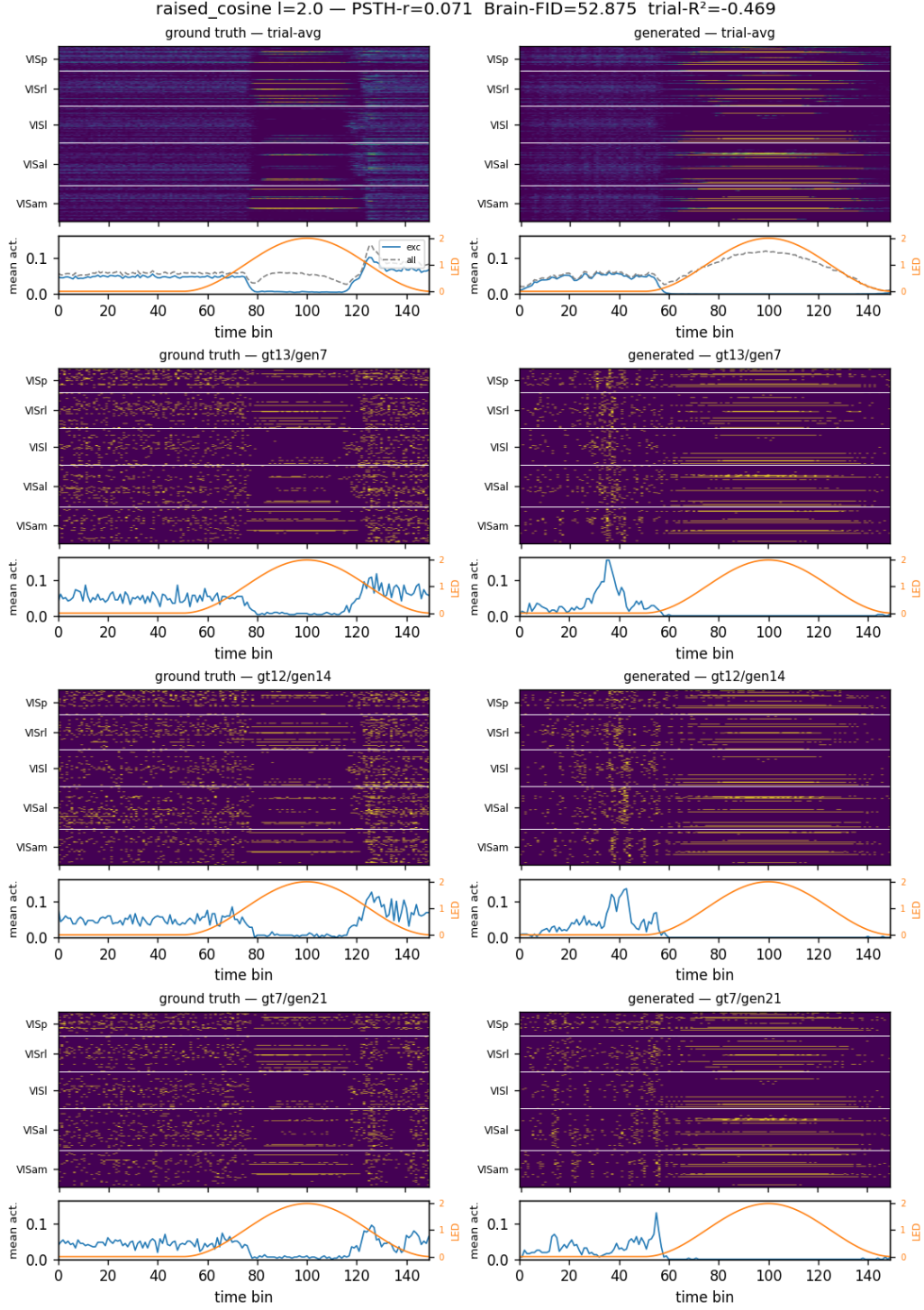


Figure 5: Qualitative perturbation prediction (lowBio, Dale, rank 2, $\tau_0 = 2$, opto scale 1.0, raised-cosine $l = 2.0$). Left: recorded; right: generated. Top row: trial-averaged population activity (blue) overlaid on the injected light waveform (orange); lower rows: example single trials with per-area rasters. The generated mean follows the light, while single-trial details diverge.